Chemistry - Undergraduate Level Assessment 14 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Which nuclide has an NMR frequency of 115.5 MHz in a 20.0 T magnetic field?
   (a) 17 O
   (b) 19 F
   (c) 29 Si
   (d) 31 P
2. Cobalt-60 is used in the radiation therapy of cancer and can be produced by bombardment of cobalt-59 with which of the following?
   (a) Neutrons
   (b) Alpha particles
   (c) Beta particles
   (d) X-rays
3. The normal modes of a carbon dioxide molecule that are infrared-active include which of the following? I. Bending II. Symmetric stretching III. Asymmetric stretching
   (a) I only
   (b) II only
   (c) III only
   (d) I and III only
4. Calculate the magnetic field responsible for the polarization of 2.5 x $10^-6 for 13C at 298K$.
   5. 0.5 T
   6. 1.2 T
   7. 2.9 T
   8. 100 T

The T_2 of an NMR line is 15 ms. Ignoring other sources of linebroadening, calculate its linewidth.

(a) 0.0471 Hz
(b) 21.2 Hz
(c) 42.4 Hz
(d) 66.7 Hz

True / False (10 marks)

Answer True or False

6. True or False: The atomic radii of the lanthanide elements decrease across the period from La to Lu due to increasing nuclear charge.

7. True or False: The NMR frequency of 31 P in a 20.0 T magnetic field is 239.2 MHz.

8. True or False: HCl and NaOH are classified as a conjugate acid-base pair in acid-base chemistry.

9. True or False: The 13 C spectrum of hexane exhibits lines with three distinct chemical shifts due to its symmetrical structure.

10. True or False: At standard temperature and pressure, Br_2 is a colorless gas.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the impact of diluting a buffer solution, composed of equal concentrations of a weak acid and its conjugate base, on its pH. Provide a detailed analysis to support your explanation.

12. Discuss the factors that contribute to the lattice enthalpy of ionic compounds, and analyze why magnesium oxide (MgO) has the greatest lattice enthalpy among typical ionic substances.

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